

Referee’s report of IEEE TAI manuscript #TAI-2026-Apr-A-00756

Summary

This paper studies open-world dense segmentation, where a deployed segmentation model may encounter image regions that do not belong to any of the known training classes. The authors argue that unknown handling in dense prediction should not be treated only as post-hoc pixel-level uncertainty thresholding. Instead, the model should produce coherent abstention regions as part of its structured segmentation output. To this end, the paper formulates Structured Abstention Segmentation (SASeg), introduces native unknown query slots in a Transformer-based region parser, and adds a prototype-sphere support module for judging whether candidate regions fall inside the support of known classes. The paper also proposes two evaluation protocols: A1 for seen-unknown structured abstention and A2 for unseen-unknown detection generalization. Experiments are reported on Synapse, AMOS22, PASCAL VOC, and Cityscapes.

I find the main idea interesting and timely. The distinction between detecting unknown pixels and structuring them into usable unknown masks is important, and the paper provides useful evidence that high pixel-level AUROC does not necessarily imply useful structured abstention. At the same time, the current empirical results also show important limitations: the method relies on seen-unknown supervision, the A2 structured outputs remain nearly unusable, and the inference procedure still depends heavily on thresholding. These issues should be clarified before the contribution can be properly assessed.

Major comment

- The supervision setting and the required training data for the unknown class should be stated more explicitly. The method requires annotations for seen unknown classes, which are collapsed into a generic unknown mask during training. This is a reasonable setting, but it means SASeg is better described as supervised structured abstention with unseen-unknown generalization, rather than generic unknown discovery. The

paper should discuss how such seen-unknown masks would be obtained in realistic applications and provide the exact class splits more prominently.

- The A2 results reveal a severe limitation that should be reflected more carefully in the claims. Across all four datasets, the detection scores can be high while the unknown F1 and IoU are almost zero. The region-level candidate analysis usefully shows that candidate purity is the bottleneck, but it also shows that the method does not yet solve structured abstention for truly unseen unknowns. The paper should present A2 primarily as a diagnostic setting, or provide a stronger method for A2 mask generation.
- The paper criticizes post-hoc thresholding, but the final inference procedure still relies heavily on thresholded scores. In A1, the decision thresholds support and unknown-query responses; in A2, it thresholds an ensemble anomaly score. Since the central message is that thresholding detection scores is insufficient for structured abstention, the authors should evaluate alternatives that more directly optimize structured mask quality. For example, RankSeg-style Dice/IoU-oriented decoders (<https://github.com/rankseg/rankseg>) could help test whether the poor A2 F1/IoU is mainly caused by the post-hoc thresholding pipeline, or instead reflects the intrinsic difficulty of generating clean unseen-unknown candidates. Other useful comparisons could include Mask2Former- or Mask DINO-style $K + 1$ baselines, region-level proposal ranking, and simple mask-refinement methods.
- The baseline results should be discussed more carefully. In Table I, the independently trained per-pixel $K + 1$ segmenter obtains an unknown F1 of 0.797 on Synapse A1, whereas SASeg obtains 0.721. This suggests that reserving an explicit unknown output space may be more important than the specific query-based architecture or prototype-sphere module. The authors should either reposition the contribution accordingly or provide stronger controlled comparisons, such as query-based $K + 1$ and MaskFormer-style $K + 1$ baselines.

Minor comments

- Please provide a clear table of the known, seen-unknown, and unseen-unknown class splits for each dataset, and describe precisely how the binary unknown training mask is constructed from the seen-unknown classes.

- Some implementation and inference details should be easier to follow, including the supervision of the background/no-object queries, the distinction between the “Support” path and U_{union} , and the validation procedure used to choose the A2 ensemble weights.
- More qualitative visualizations would be helpful, especially for A2. Showing successful and failed unknown-query candidates would make the detection–structuring gap much more concrete.
- The computational cost of SASeg should be reported, including training time, inference time, memory usage, and the overhead of the prototype-sphere support module relative to simpler $K + 1$ segmentation baselines.